

ML4GW+Controls Workshop

LVK meeting - Pune 2026

Welcome - Hopefully you had a productive LVK meeting!

Agenda for the day

- Morning session I (09:30-10:45): Bhaskara 3
 - Intro to neural nets and basics on function and distribution approximation
- Coffee break
- Morning session II (11:15-12:45): Bhaskara 3
 - Building a prototype search and inference model using the ML4GW framework
- Lunch break
- Afternoon session 1 (14:15-15:15) - Keynote lecture: Bhaskara 3
 - [Keynote talk](#) by Prof. Ajit Kembhavi
- Coffee break; transition to TLC classroom
- Afternoon session II (15:30-18:00)- TLC smart classroom
 - AI for controls: group activity

Use of ML in LVK production data analysis

- *Neural networks now routinely discover and characterize CBCs!*
- **Aframe** is the first NN-based BBH search deployed live in O4
 - Deployed alongside other matched-filter searches
- Rapid parameter estimation **AMPLFI** deployed alongside Aframe to generate skymaps, chirp-mass estimates, other source properties required for alerts
- Among burst searches, MLy is a NN-based search deployed alongside CWB

Aframe + AMPLFI deployed in O4

First ever BBH discoveries using NNs: 23 BBHs discovered in O4

Events from first week: [S250904cv](#), [S250904br](#), [S250904ae](#), [S250901cb](#), [S250830bp](#), [S250830m](#)

GCN Circular 41606

Subject LIGO/Virgo/KAGRA S250830bp: Identification of a GW compact binary merger candidate
Event [LIGO/Virgo/KAGRA S250830bp](#)
Date 2025-08-30T10:55:06Z (4 days ago)
From Pedro Jeronimo Santos da Silva <pedro.jeronimo@unesp.br>
Via Web form

The LIGO Scientific Collaboration, the Virgo Collaboration, and the KAGRA Collaboration report:

We identified the compact binary merger candidate S250830bp during real-time processing of data from LIGO Hanford Observatory (H1), LIGO Livingston Observatory (L1), and Virgo Observatory (V1) at 2025-08-30 10:24:18.852 UTC (GPS time: 1440584676.852). The candidate was found by the Aframe [1], cWB [2], cWB BBH [3], GstLAL [4], MBTA [5], MLy [6], PyCBC Live [7], and SPIIR [8] analysis pipelines.

S250830bp is an event of interest because its false alarm rate, as estimated by the online analysis, is $3.2e-10$ Hz, or about one in $1e2$ years. The event's properties can be found at this URL:

Two sky maps are available at this time and can be retrieved from the GraceDB event page:

* [amplfi.muliorder.fits,0](#), an initial localization generated by AMPLFI [10], distributed via GCN and SCiMMA notices about 24 seconds after the candidate event time.

* [bayestar.muliorder.fits,0](#), an initial localization generated by BAYESTAR [11], distributed via GCN and SCiMMA notices about 5 minutes after the candidate event time.

Challenge with ML research in GW Science

Common ML Frameworks are mature

- Math operations, array manipulations etc. are well supported on co-processors like GPUs.
- Core components of building neural networks exist

Lack of Domain-level adaptations

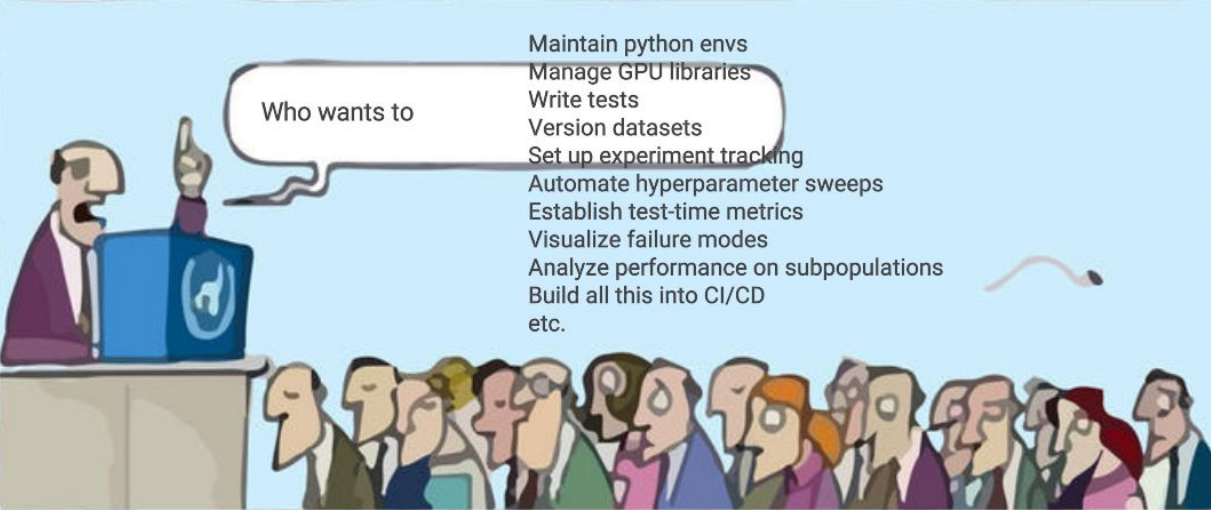
- Things that happen around training a neural network.
- Data pre-processing, CPU $\leftrightarrow$ GPU data transfer etc.
- Simulations using co-processors.

Fast implementation of ideas, testing algorithms on vast amounts of data for production readiness can be difficult.



Who wants to train ML models?

Standalone
analysis, e.g. for a
paper, lack of
integration testing
does not easily port
over to production.



Who wants to

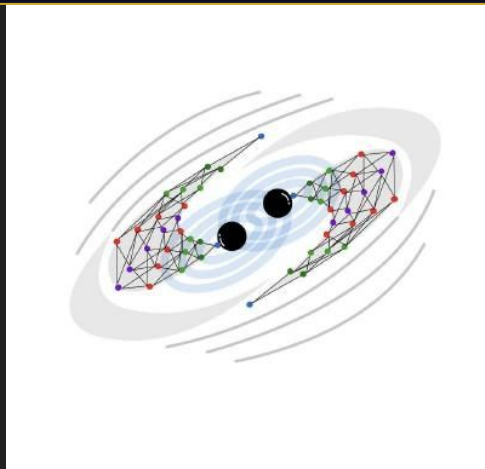
- Maintain python envs
- Manage GPU libraries
- Write tests
- Version datasets
- Set up experiment tracking
- Automate hyperparameter sweeps
- Establish test-time metrics
- Visualize failure modes
- Analyze performance on subpopulations
- Build all this into CI/CD
- etc.

Credits: MW Coughlin

ML4GW software stack

Tools to help GW data analysis on
accelerated hardware

github.com/ML4GW



<https://joss.theoj.org/papers/10.21105/joss.08836>

Morning sessions

Session 1

- Basics of neural networks: why they work?
 - Function and distribution approximation property
 - Toy problem: linear regression revisited using likelihood-free inference

Session 2

- Features of the ml4gw software stack
 - Batched data preparation for training at scale
- We will build a prototype like Aframe and AMPLFI

(hopeful) Takeaway: Accelerated GW *data preparation* for your ML research